

Project Initialization and Planning Phase

Date	30 June 2026
Team ID	xxxxxx
Project Title	Rising Waters: A Machine Learning Approach to Flood Prediction
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposed solution aims to improve flood prediction using machine learning techniques. It analyzes historical weather and rainfall data to predict flood risk accurately and provide early warnings. The system integrates the best-performing machine learning model into a Flask web application, enabling disaster management authorities to make timely decisions and improve emergency response.

Project Overview	
Objective	Develop an intelligent flood prediction system using Machine Learning to analyze historical weather and rainfall data and accurately predict the likelihood of flood occurrence. The system aims to provide early warnings that enable disaster management authorities to take timely preventive actions and minimize the impact of floods.
Scope	The project focuses on predicting flood risk using meteorological parameters such as annual rainfall, seasonal rainfall, cloud cover, humidity, temperature, and other weather-related features. Multiple classification algorithms including Decision Tree, Random Forest, K-Nearest Neighbors (KNN), and XGBoost are trained and evaluated to identify the most accurate prediction model. The final model is integrated into a Flask web application for real-time flood risk prediction.
Problem Statement	
Description	Floods are among the most destructive natural disasters, causing significant loss of life, damage to infrastructure, and economic disruption every year. Existing flood forecasting systems often fail to provide timely and accurate predictions due to the complexity of weather patterns and limitations of traditional prediction techniques. This creates challenges for disaster management authorities in issuing early warnings and planning effective emergency responses.
Impact	An accurate flood prediction system can significantly improve disaster preparedness by providing early warnings before flood events occur. It helps authorities make informed decisions, reduce human casualties, protect infrastructure, optimize resource allocation, and improve overall emergency response efficiency.
Proposed Solution	

Approach	The proposed solution utilizes supervised machine learning techniques to analyze historical weather data and identify flood-prone conditions. Data pre-processing, feature engineering, and classification algorithms are employed to build predictive models. After comparing multiple algorithms, the best-performing model is deployed through a Flask web application that provides instant flood risk predictions based on user-input weflood prediction
Key Features	- Implementation of a machine learning-based flood prediction model.

Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CPU/GPU specifications, number of cores	T4 GPU
Memory	RAM specifications	8 GB
Storage	Disk space for data, models, and logs	1 TB SSD
Software		
Frameworks	Python frameworks	Flask
Libraries	Additional libraries	scikit-learn, pandas, numpy, matplotlib, seaborn
Development Environment	IDE	Jupyter Notebook, pycharm
Data		
Data	Source, size, format	Flood Risk in India Dataset (Kaggle), CSV Format